NutriSync Project Execution Guide

This document explains how to execute and run all services required for the NutriSync project. The execution script starts the backend services in the correct sequence, including the FastAPI nutrition engine, Google Fit ingestion service, and Node.js backend server.

Execution Script

```
#!/usr/bin/env bash

echo "Starting NutriSync services..."

# Start Python engine
cd nutricalc-engine
uvicorn app.main:app --host 0.0.0.0 --port 8000 &

# Google Fit ingestion
python google_fit_ingestion.py &

cd ..

# Start Node backend (also serves frontend build)
node server.js
```

Step-by-Step Execution Flow

Step	Description
1	The bash script begins execution and displays a startup message: 'Starting NutriSync services...'
2	The script enters the 'nutricalc-engine' directory using the cd command.
3	The FastAPI-based nutrition calculation engine is launched using Uvicorn on port 8000.
4	The Google Fit ingestion service is started in the background using Python.
5	The script moves back to the root project directory using 'cd ..'.
6	The Node.js backend server is started using 'node server.js'. This backend also serves the frontend build.

Services Started by the Script

Service	Purpose
FastAPI Nutrition Engine	Handles nutrition-related calculations and API operations.
Google Fit Ingestion Service	Collects and processes Google Fit health data.
Node.js Backend	Serves backend APIs and frontend build files.

Important Notes

- The '&' operator runs services in the background so that multiple services can execute simultaneously.
- Ensure Python, Node.js, and all required dependencies are installed before execution.
- Uvicorn must be installed in the Python environment.
- The backend server depends on server.js being present in the project root directory.
- The FastAPI engine runs on port 8000 by default.